

K-Means Clustering

Introduction to K-Means Clustering

K-Means Clustering is one of the most popular **unsupervised machine learning algorithms** for grouping similar data points into clusters.

The main idea is:

“Similar data points should belong to the same group (cluster).”

It is widely used in:

- Customer segmentation
- Recommendation systems
- Image compression
- Student performance grouping
- Market analysis
- Pattern recognition

What is Clustering?

Clustering means dividing data into groups based on similarity.

Example:

Suppose a college has student data:

- CGPA
- Attendance
- Skills
- Projects

The college wants to divide students into:

- High performers
- Average performers
- Low performers

K-Means automatically creates these groups.

What Does “K” Mean?

In K-Means:

- **K = Number of clusters/groups**

Example:

- K = 2 → 2 clusters
- K = 3 → 3 clusters
- K = 4 → 4 clusters

The user decides the value of K before training.

Important Terminologies

1. Cluster

A group of similar data points.

2. Centroid

The center point of a cluster.

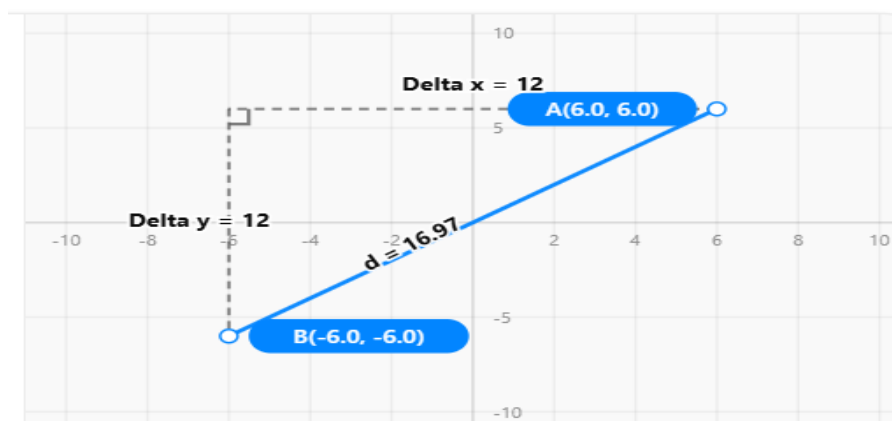
3. Distance

Used to measure similarity between points.

Usually, Euclidean Distance is used.

Distance Formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$



Smaller distance means more similarity.

Working of K-Means Algorithm

The K-Means algorithm works in multiple steps.

Step 1: Choose Number of Clusters (K)

First decide how many clusters are needed.

Example:

- $K = 3$

Means:

- Cluster 1
- Cluster 2
- Cluster 3

Step 2: Initialize Centroids Randomly

The algorithm randomly selects K points as initial centroids.

Example:

If $K = 3$,
then 3 random points are selected.

These points act as temporary centers.

Step 3: Calculate Distance

For every data point:

- Calculate distance from each centroid
- Assign the point to the nearest centroid

Example:

A point is nearest to Cluster 2 centroid,
so it becomes part of Cluster 2.

Step 4: Create Clusters

After assigning all points:

- Groups/clusters are formed.

Each cluster now contains similar data points.

Step 5: Recalculate Centroids

Now calculate the new centroid of each cluster.

The centroid is the mean of all points in that cluster.

Centroid Formula:

$$Centroid = \left(\frac{\sum x}{n}, \frac{\sum y}{n} \right)$$

Step 6: Repeat the Process

Again:

- Calculate distances
- Reassign points
- Update centroids

This process continues until:

- Centroids stop changing.

This is called **convergence**.

Final Output

Finally:

- Similar data points are grouped together.
- Final clusters are produced.

Example of K-Means

Suppose a shopping mall has customer data:

- Age
- Income

K-Means can divide customers into:

1. High-income customers
2. Medium-income customers
3. Low-income customers

This helps businesses in marketing.

Real-World Applications of K-Means

1. Customer Segmentation

Grouping customers based on buying behavior.

2. Recommendation Systems

Netflix and YouTube recommend similar content.

3. Image Compression

Reducing image size by clustering colors.

4. Medical Diagnosis

Grouping patients with similar symptoms.

5. Education Systems

Grouping students by performance.

Advantages of K-Means

- Simple and easy to understand
- Fast algorithm
- Works well on large datasets
- Efficient for clustering tasks

Disadvantages of K-Means

- Need to specify K manually
- Sensitive to outliers
- Different initial centroids may produce different results
- Works best with spherical clusters

What is WCSS?

WCSS stands for:

Within Cluster Sum of Squares

It measures:

How compact the clusters are.

Formula:

$$WCSS = \sum (x_i - c)^2$$

Where:

- x_i = data point
- c = centroid

Lower WCSS means better clustering.

Elbow Method

The Elbow Method helps find the best value of K.

Steps of Elbow Method

1. Train K-Means for different K values
2. Calculate WCSS for each K
3. Plot graph:
 - X-axis → Number of clusters (K)

- Y-axis → WCSS

4. Find the “elbow point”

The elbow point is the best K value.

Why Called “Elbow”?

Because the graph looks like a human arm elbow.

At this point:

- WCSS decreases slowly after that.
- Adding more clusters gives little improvement.

K-Means Workflow Summary

1. Choose K
2. Initialize centroids
3. Calculate distances
4. Assign clusters
5. Update centroids
6. Repeat until convergence

Interview Questions

Q1. Is K-Means supervised or unsupervised?

Answer:

Unsupervised learning algorithm.

Q2. What is the role of centroid?

Answer:

Centroid represents the center of a cluster.

Q3. Why do we use the Elbow Method?

Answer:

To find the optimal value of K.

Q4. What is convergence in K-Means?

Answer:

When centroids stop changing.

Conclusion

K-Means Clustering is a powerful and simple unsupervised learning algorithm used for grouping similar data points into clusters. It is widely used in real-world applications like customer segmentation, recommendation systems, education analytics, and image processing. The algorithm works by repeatedly assigning points to the nearest centroid and updating cluster centers until stable clusters are formed.